Functions on $\mathbb{R}$

$I = (\alpha, \beta)$: an open interval

$\lim_{x\to a} f(x) = L$: def. $\forall \varepsilon > 0 .\ \exists \delta > 0 .\ \forall x \in I .\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

$f : I \setminus \{a\} \to \mathbb{R}$

Ex. Show $\lim_{x\to a} (mx+b) = ma+b$

$\forall \varepsilon > 0 .\ \exists\ \delta = \frac{\varepsilon}{|m|}$

$0 < |x-a| < \delta \Rightarrow |mx+b-(ma+b)| = |m||x-a| < \varepsilon$.

$\Rightarrow \lim_{x\to a} (mx+b) = ma+b$

Ex. $\lim_{x\to a} x^2 = a^2$

$|x^2 - a^2| = |x-a||x^2 + ax + a^2|$

Let $M : |a| + 1$ — non-negative

$\forall \varepsilon > 0 .\ \exists\ \delta = \min\{1, \frac{\varepsilon}{|M^2 + aM + a^2|}\}$.

$|x-a| < \delta \Rightarrow |x| - |a| \le |x-a| < \delta \le 1$.

$|x| < |a| + 1 < M$.

$|x-a| < \delta \Rightarrow |x^2 - a^2| = |x-a||x^2 + ax + a^2|$

$< \delta \cdot |M^2 + aM + a^2|$

$< \frac{\varepsilon}{|M^2 + aM + a^2|} \cdot |M^2 + aM + a^2| = \varepsilon$

$\therefore \lim_{x\to a} x^2 = a^2$

$|x^3 - a^3| = |x-a||x^2 + ax + a^2|$

Let $M : |a| + 1$

$\forall \varepsilon > 0 .\ \exists\ \delta = \min\{1, \frac{\varepsilon}{|M^2 + aM + a^2|}\}$.

$|x-a| < \delta \Rightarrow |x| - |a| \le |x-a| < \delta \le 1$.

$|x| < |a| + 1 < M$.

$|x-a| < \delta \Rightarrow |x^3 - a^3| = |x-a||x^2 + ax + a^2|$

$< \delta \cdot |M^2 + aM + a^2|$

$< \frac{\varepsilon}{|M^2 + aM + a^2|} \cdot |M^2 + aM + a^2| = \varepsilon$

$\therefore \lim_{x\to a} x^3 = a^3$

remk: if $f(x) = g(x)$, $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$

Ex. $g(x) = \frac{x^3 x^3 - x - 1}{x^3 - 1}$, $\lim_{x\to 1} g(x)$.

$g(x) = \frac{(x^3-1)(x+1)}{(x^3-1)} = x+1$.

$\lim_{x\to 1} g(x) = \lim_{x\to 1} (x+1) = 2$.

The definition can be summarized as

$\lim_{x\to a} f(x) = L \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I :\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Its negation is

$\lim_{x\to a} f(x) \neq L \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in I :\ 0 < |x-a| < \delta$ and $|f(x) - L| \ge \varepsilon_0$.

Sequential Criterion

Thm 2.6: $\lim_{x\to a} f(x) = L \iff$ $\forall$ $\{x_n\}_{n=1}^{\infty}$ with $x_n \neq a$ $\forall n \in \mathbb{N}$ and $x_n \to a$, $f(x_n) \to L$

Proof: ⇒ If $\lim_{x\to a} f(x) = L$, then $\forall \varepsilon > 0 .\ \exists \delta > 0 .\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

If $x_n \to a$, $\exists N \in \mathbb{N} .\ n \ge N \Rightarrow 0 < |x_n - a| < \delta$

Hence $|f(x_n) - L| < \varepsilon \Rightarrow f(x_n) \to L$.

⇐ Suppose $\lim_{x\to a} f(x) \neq L$. Then exists $\varepsilon_0 > 0 :$ $\forall \delta > 0 .\ \exists x \in I \setminus \{a\}$ $|x-a| < \delta$ and $|f(x) - L| \ge \varepsilon_0$.

Choose $\delta = 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots$ such that $0 < |x_n - a| < \frac{1}{n}$ and $|f(x_n) - L| \ge \varepsilon_0$.

For each $n \in \mathbb{N}$, $\delta_n = \min\{\frac{1}{n}, \frac{a}{2}\} > 0$.

$\Rightarrow |x_n - a| < \delta_n$ $|f(x_n) - L| \ge \varepsilon_0$.

which construct a sequence $x_n \to a$ but $f(x_n)$ doesn't converge to L.

contrapositive : "$p \Rightarrow q$ ⇔ $\neg q \Rightarrow \neg p$"

Theorem 2.8

I is an open interval containing a. f, g are function on I.

$\lim_{x\to a} f(x) = L$. $\lim_{x\to a} g(x) = M$.

Then

① $\lim_{x\to a} (f+g)(x) = L+M$.

② $\lim_{x\to a} (cf)(x) = cL$.

③ $\lim_{x\to a} (fg)(x) = LM$.

④ $\lim_{x\to a} (\frac{1}{g})(x) = \frac{1}{M}$ given $M \neq 0$

Proof: ① any sequence in $I \setminus \{a\}$ with $x_n \to a$.

By thm 2.6. $f(x_n) \to L$. $g(x_n) \to M$

By algebraic rule for Sequence :

$f(x_n) + g(x_n) \to L+M$.

Since it holds for any Sequence. By thm 2.6 $\lim_{x\to a} (f+g)(x) = L+M$.

②③④ can be proved Similarly

2.9 Squeeze theorem

① if $g(x) \le h(x) \le f(x)$ for all $x \in I \setminus \{a\}$

and $\lim_{x\to a} g(x) = \lim_{x\to a} f(x) = L$.

then $\lim_{x\to a} h(x) = L$

② if $|g(x)| \le M$ for all $x \in I \setminus \{a\}$ and $f(x) \to 0$ for $x \to a$.

then $h(x) \to 0$ as $x \to a$

2.10 Theorem for Comparison

Suppose $a \in \mathbb{R}$. f, g have limits as $x \to a$

and $f(x) \le g(x)$ for all $x \in I \setminus \{a\}$.

then $\lim_{x\to a} f(x) \le \lim_{x\to a} g(x)$

One-sided Limit and Limit at Infinity

$\forall \varepsilon > 0 .\ \exists \delta > 0$, such that $a < x < a+\delta \in I .\ a < x < a+\delta \Rightarrow |f(x) - L| < \varepsilon$.

then we say $\lim_{x\to a^+} f(x) = L$: $\lim_{x\to a^+} f(x)$

$\forall \varepsilon > 0 .\ \exists \delta > 0$, such that $a-\delta \in I .\ a-\delta < x < a \Rightarrow |f(x) - L| < \varepsilon$.

then we say $f(x) \to L$: $\lim_{x\to a^-} f(x) = L$

Ex. Prove: $f(x) = \begin{cases} x+1, & x > 0 \\ x-1, & x < 0 \end{cases}$

has one-sided Limit as $x \to 0$ but $\lim_{x\to 0} f(x)$ does not exist

① $\forall \varepsilon > 0\ \exists \delta = \varepsilon .\ 0 < x < \delta \Rightarrow 1 \le x+1 < 1+\delta \Rightarrow |x+1 - 1| < \delta < \varepsilon$.

$\therefore \lim_{x\to 0^+} f(x) = 1$.

② $\forall \varepsilon > 0\ \exists \delta = \varepsilon .\ -\delta < x < 0 \Rightarrow |x-1-(-1)| = |x| < \delta < \varepsilon$.

$\therefore \lim_{x\to 0^-} f(x) = -1$.

③ Let $x_n = \frac{(-1)^n}{n} \to 0$. $f(x_{2n}) = \frac{(-1)^n}{n} + (-1)^n = (-1)^n (1 + \frac{1}{n})$

$\Rightarrow f(x_n)$ does not converge. $\Rightarrow \lim_{x\to 0} f(x)$ doesn't exist

Ex. prove: $\lim_{x\to 0} \sqrt{x} = 0$

$\forall \varepsilon > 0\ \exists \delta = \varepsilon^2 .\ 0 < x < \delta \Rightarrow |\sqrt{x}| = \sqrt{|x|} < \varepsilon$

$\Rightarrow \lim_{x\to 0} \sqrt{x} = 0$

Theorem 2.14

$\lim_{x\to a} f(x)$ exists and equals to L if and only if $L = \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

Proof: ⇒ if $\lim_{x\to a} f(x) = L$ as $x \to a$. given $\varepsilon > 0$, choose $\delta > 0$

$0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Since $a < x < a+\delta \Rightarrow a-\delta < x < a$ both satisfy $0 < |x-a| < \delta$.

Hence both one-side limit exists and equals to L

⇐ if $L = \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$

$a < x < a+\delta, \Rightarrow |f(x) - L| < \varepsilon$.

and $a-\delta < x < a \Rightarrow |f(x) - L| < \varepsilon$.

See $\delta = \min\{\delta_1, \delta_2\}$.

then $|x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

Thus $\lim_{x\to a} f(x) = L$ as $x \to a$.

3.25 Definition

Let $a, L \in \mathbb{R}$, and let f be a real function.

(i) The function $f(x)$ is said to *converge to L as $x \to +\infty$* if and only if there is a $c > 0$ such that $(c, +\infty) \subseteq \mathrm{Dom}(f)$ and, given $\varepsilon > 0$, there is an $M > c$ such that

$$x > M \implies |f(x) - L| < \varepsilon.$$

In this case, we write

$$\lim_{x\to+\infty} f(x) = L \quad \text{or} \quad f(x) \longrightarrow L \ \text{ as } x \to +\infty.$$

Similarly, $f(x)$ is said to *converge to L as $x \to -\infty$* if and only if there is a $c > 0$ such that $(-\infty, -c) \subseteq \mathrm{Dom}(f)$ and, given $\varepsilon > 0$, there is an $M < -c$ such that

$$x < M \implies |f(x) - L| < \varepsilon.$$

We write

$$\lim_{x\to-\infty} f(x) = L \quad \text{or} \quad f(x) \longrightarrow L \ \text{ as } x \to -\infty.$$

(ii) The function $f(x)$ is said to *converge to $+\infty$ as $x \to a$* if and only if there is an open interval I containing a such that $I \setminus \{a\} \subseteq \mathrm{Dom}(f)$ and, given $M \in \mathbb{R}$, there is a $\delta > 0$ such that

$$0 < |x-a| < \delta \implies f(x) > M.$$

We write

$$\lim_{x\to a} f(x) = +\infty \quad \text{or} \quad f(x) \longrightarrow +\infty \ \text{ as } x \to a.$$

Similarly, $f(x)$ is said to *converge to $-\infty$ as $x \to a$* if and only if there is an open interval I containing a such that $I \setminus \{a\} \subseteq \mathrm{Dom}(f)$ and, given $M \in \mathbb{R}$, there is a $\delta > 0$ such that

$$0 < |x-a| < \delta \implies f(x) < M.$$

We write

$$\lim_{x\to a} f(x) = -\infty \quad \text{or} \quad f(x) \longrightarrow -\infty \ \text{ as } x \to a.$$

Other infinite limit combinations

The remaining combinations are defined analogously: $f(x)$ may tend to $+\infty$ or $-\infty$ as $x \to a+$ or $x \to a-$, and it may tend to $+\infty$ or $-\infty$ as $x \to +\infty$ or $x \to -\infty$.

Ex. prove: $1/x \to 0$ as $x \to \infty$

$\forall \varepsilon > 0 \quad \exists\ M = \frac{1}{\varepsilon} \quad x > M \Rightarrow 0 < \frac{1}{x} - 0 < \frac{1}{M} < \varepsilon$

Ex. $f(x) = \frac{3x^2 - 2}{2x^2 - 3x + 1}$

prove $\lim_{x\to 1^+} f(x) = -\infty$.

$f(x) = \frac{3x^2 - 2}{(2x-1)(x-1)}$

$\exists\ \delta \in (0, 1) .\ 1 - \delta < x < 1 \Rightarrow \frac{1}{M} < x^2$. $3x + 1 < 0$.

$2x - 1 < 0$

Continuity

Def: f is continuous at a if: $\forall \varepsilon > 0 .\ \exists\ \delta > 0$ such that $x \in I .\ |x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$

f is continuous on I if: it is continuous at every points of I.

$\lim_{x\to a} f(x) = f(a)$

Endpoint convention. If $f : [a, b] \to \mathbb{R}$, continuity at the endpoints is understood relative to the domain:

$$\lim_{x\to a^+} f(x) = f(a), \qquad \lim_{x\to b^-} f(x) = f(b).$$

Theorem 3.31: Sequential Characterization

f is cont. at $a \Leftrightarrow$ every $\{x_n\} \subseteq I$ with $x_n \to a$. Satisfies $f(x_n) \to f(a)$.

proof: ⇒ If f is cont. at $a \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0 .\ 0 < |x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$.

Since $x_n \to a$. choose this δ. $\exists N .\ n > N \Rightarrow |x_n - a| < \delta \Rightarrow |f(x_n) - f(a)| < \varepsilon$

$\Rightarrow f(x_n) \to f(a)$ as $n \to \infty$

⇐ Assume f is not cont. at a.

$\exists\ \varepsilon_0 > 0\ \forall \delta > 0 .\ \exists x \in I .\ 0 < |x-a| < \delta \Rightarrow |f(x) - f(a)| \ge \varepsilon_0$.

for every $n \in \mathbb{N}$. Let $\delta = \frac{1}{n}$, $\exists x_n \in I .\ 0 < |x_n - a| < \frac{1}{n}$. Constitute $x_n \to a$.

But $|f(x_n) - f(a)| \ge \varepsilon_0$. $f(x_n)$ does not converge to $f(a)$

$\neg q \Rightarrow \neg p$ ⇔ $p \Rightarrow q$

Theorem : Algebra of Cont. functions

① $f, g : I \to \mathbb{R}$ be Cont. at $a \in I$

Then $f+g$. $f \cdot g$. $\frac{f}{g}$ $(g(a) \neq 0)$ are Cont. at a.

② if f, g Cont. on I

Then $f+g$. $f \cdot g$. $\frac{f}{g}$ $(g(x) \neq 0)$ are Cont. on I.

proof: $\lim_{x\to a} (f+g)(x) = f(a) + g(a) = (f+g)(a)$

Def: composition

$f : A \to \mathbb{R}$. $g : B \to \mathbb{R}$. assume : $f(A) \subseteq B$.

The Composition of g with f : $g \circ f : A \to \mathbb{R}$. $(g \circ f)(x) := g(f(x))$

Theorem: Continuity & Composition

① $f : A \to \mathbb{R}$. $g : B \to \mathbb{R}$. if f is Cont at $a \in A$, g is Cont. at $f(a) \in B$.

the $g \circ f$ is Cont. at a.

② if f is cont on A & g is Cont. on B. Then $g \circ f$ is cont. on A.

proof: Let $x_n \in A$ and $x_n \to a$. cont of f $f(x_n) \to f(a)$.

since $f(a) \in B$, $f(x) \in B\ \forall n$ cont of g $g(f(x_n)) \to g(f(a))$.

Hence $g \circ f$ is cont. at a.

Def. Bounded Function

$E \subseteq \mathbb{R}$, $f : E \to \mathbb{R}$, f is bounded if $f(E) = \{f(x) : x \in E\}$ is bounded

Equivalently, there is an $M > 0$ such that $|f(x)| \le M$.

in this case f is said to be dominated by M on E.

Ex. $f : [a, +\infty) \to \mathbb{R}$, $f(x) = \frac{2x}{1+x^2}$. For $x \in [a, +\infty)$.

$|f(x)| < \frac{2|x|}{1+x^2} = \frac{2|x|}{|x|^2} < \frac{2}{|x|} \le \frac{2}{a} = g$

$\Rightarrow f$ is bounded on $[a, +\infty)$. (f is dominated by g on $[a, +\infty)$)

Theorem: Extreme Value Theorem

① $I = [a, b]$. $f : I \to \mathbb{R}$ be Cont Then f is bounded on I.

② Moreover if $M = \sup_{x\in I} f(x)$. $m = \inf_{x\in I} f(x)$.

Then there are points $x_M, x_m \in I$:

$f(x_M) = M$. $f(x_m) = m$

proof: Suppose f is unbounded. For each $n \in \mathbb{N}$ choose $x_n \in I$ such that $|f(x_n)| > n$.

since $x_n \in I$. (x_n) is bounded. $\xrightarrow{\text{B-W}}$ (x_n) has a Convergent Subseq. (x_{n_k}).

Say $x_{n_k} \to c$. since f is Continuous. $f(x_{n_k}) \to f(c) \in \mathbb{R}$.

$|f(x_{n_k})| > n \to \infty$. Hence a Contradiction.

② Let $M = \sup_{x\in I} f(x)$. Choose $x_n \in I$ such that $f(x_n) > M - \frac{1}{n}$. By thm

$f(x_n) = \lim_{k\to\infty} f(x_{n_k}) \ge M \Rightarrow f(x_n) = M$

x_m can be obtained similarly ☆ Seq $x_n \to a$ ⇒ All SubSeq $x_{n_k} \to a$

3.27 Remark (why both hypotheses matter). Neither compactness hypothesis can be omitted:

- $f(x) = 1/x$ is continuous but unbounded on the bounded, nonclosed interval $(0, 1)$;
- $f(x) = x$ is continuous and bounded on $(0, 1)$, but $\sup f((0,1)) = 1$ is not attained;
- $f(x) = x$ is continuous but unbounded on the closed, unbounded interval $[0, +\infty)$.

① closed bounded interval $I = [a, b]$

② continuous $f : I \to \mathbb{R}$

Lemma: Persistence of positivity

$a < b$. $f : [a, b] \to \mathbb{R}$ if f is Continuous at x_0 and $f(x_0) > 0$.

Then there are $\alpha > 0$ and an $x_1 \in (x_0, b)$ such that $f(x) > \frac{f(x_0)}{2}$. ($x \in [x_0, x_1)$)

Proof:

See $\varepsilon = f(x_0)/2 > 0$. Since f is Cont. at x_0. Choose $\delta > 0$. such that: $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \frac{f(x_0)}{2}$

Choose $x_1 \in (x_0, b)$. close to x_0 that $x_1 - x_0 < \delta \Rightarrow \forall x \in [x_0, x_1)$ $|f(x) - f(x_0)| < \frac{f(x_0)}{2}$

$\Rightarrow f(x) > f(x_0) - \frac{f(x_0)}{2} = \frac{f(x_0)}{2} > \varepsilon$

$\exists\ \delta\ |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \frac{f(x_0)}{2} = \varepsilon$

Choose x_1 so close to x_0 that $|x_1 - x_0| < \delta$

$\Rightarrow \forall x \in [x_0, x_1)$ $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \frac{f(x_0)}{2}$

$\Rightarrow f(x) > \frac{f(x_0)}{2} > \varepsilon$

Intermediate Value Theorem

$f : [a, b]$ be Continuous. if y_0 lies strictly between $f(a)$ and $f(b)$. then there is $x_0 \in (a, b)$. $f(x_0) = y_0$.

proof: Assume : $f(a) < y_0 < f(b)$. Set $E = \{x \in [a, b] : f(x) < y_0\}$. $x_0 := \sup E$. Completeness Axiom : $\sup E \in \mathbb{R}$. Let $\sup E = \xi$!

E is nonempty & bounded. Choose $y_0 \in E$ with $x_0 \ge x_0$. since $E \subseteq [a, b] \subseteq \mathbb{R}$. $x_0 \in [a, b]$

and since f is cont. $f(x_0) = \lim_{x\to x_0} f(x) < y_0$. also shows $x_0 < b$

$\exists \sup E \in \mathbb{R}$ since $x_0 \in E \subseteq [a, b]$

If $f(x_0) < y_0$. Let $h(x) = y_0 - f(x)$. $h(x)$ is Cont. and positive at x_0.

By Lemma : $\exists\ \varepsilon > 0 .\ x_0 \in (x_0, x_0 + \delta) .\ h(x) > \varepsilon$ ($\forall x \in [x_0, x_1)$)

$\Rightarrow h(x) > \varepsilon \neq 0$ $f(x) < y_0$. since $x_0 < \sup E$ & $x_1 \in E$ & $x_1 > x_0$. a Contradiction.

$\therefore f(x_0) \not< y_0$

$x_0 < b$ ⇒ $x_0 \in [a, b)$

$f(x_0) = \lim_{x\to x_0} f(x) \in \mathbb{R}$

$f(x_0) > y_0$

$f(x) < y_0 \Rightarrow \exists x_1 > x_0 .\ f(x_1) < y_0$.

Together with Extreme value theorem : IVT shows that $f([a, b])$ is a closed interval

When continuity fail ?

f is Cont. at $a \Leftrightarrow \lim_{\substack{x\to a \\ x\in I}} f(x) = f(a)$. fail : ① $\lim_{\substack{x\to a \\ x\in I}} f(x)$ does not exists

② $\lim_{\substack{x\to a \\ x\in I}} f(x) \neq f(a)$.

Ex. $f(x) = \begin{cases} |x|/x, & x \neq 0 \\ 1, & x = 0 \end{cases}$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} 1 = 1$. $\lim_{x\to 0^-} f(x) = -1$.

$\Rightarrow \lim_{x\to 0} f(x)$ does not exists $\Rightarrow f$ is not cont. at a

Ex. $f(x) = \begin{cases} \sin(1/x), & x \neq 0 \\ 1, & x = 0 \end{cases}$

Let. $x_n = \frac{1}{2n\pi} \to 0$. $f(x_n) = \sin(2n\pi) = 0$. $\forall n$.

$y_n = \frac{1}{2n\pi + \frac{\pi}{2}} \to 0$. $f(x_n) = 1$. $\forall n$.

$\lim_{x\to 0} f(x) \Rightarrow \lim_{n\to\infty} f(x_n) \Rightarrow$ limit doesn't exists

Def: Uniform Continuity

f is uniformly continuous on $I \overset{\text{def}}{\Leftrightarrow} \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in I .\ |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.

Lemma: if $f : I \to \mathbb{R}$ is uniformly cont on I then f is Continuous on I

proof: Fix $a \in I$. $\forall \varepsilon > 0\ \exists \delta > 0 .\ |x-a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$. $\Rightarrow f(x)$ is Cont. at a.

$\Rightarrow$ consequently $f(x)$ is cont. on I

Ex. $f(x) = x^2$ is uniformly Continuous on $(0, 1)$

proof: $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

$|x^2 - y^2| = |x+y||x-y| < \varepsilon$

$0 < x, y < 1 \Rightarrow 0 < x+y < 2$

take $\delta < \frac{\varepsilon}{2}$. $|x+y||x-y| < 2|x-y| < 2 \cdot \frac{\varepsilon}{2} = \varepsilon$

Hence $f(x)$ is uniformly continuous on $(0, 1)$

Ex. $f(x) = \sin(1/x)$ is not uniformly continuous on $\mathbb{R}$

equivalent to $\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in \mathbb{R} .\ |x-y| < \delta \Rightarrow |f(x) - f(y)| \ge \varepsilon_0$

for $\delta > \frac{1}{n}$, $n \in \mathbb{N}$ $\exists x_n, y_n \in \mathbb{R}$. $|x_n - y_n| < \frac{1}{n} \Rightarrow |f(x_n) - f(y_n)| \ge \varepsilon_0$

Let $x_n = \frac{1}{2n\pi + \frac{\pi}{2}}$, $y_n = \frac{1}{2n\pi}$ $|x-y| \to 0$. $|f(\sin(x) - f(y_n)| = |\sin(2n\pi + \frac{\pi}{2}) - 0| = 1 \ge \varepsilon_0$.

Unbounded

2.38 Cauchy Sequence (Lemma)

$f : I \to \mathbb{R}$ is uniformly Continuous. If $\{x_n\} \subseteq I$ is a Cauchy Sequence.

then $\{f(x_n)\}$ is a Cauchy Sequence.

proof: $\forall \varepsilon > 0\ \exists \delta > 0 .\ |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

Since $\{x_n\} \subseteq I$ is Cauchy Sequence for this δ $\exists N .\ m, n \ge N \Rightarrow |x_m - x_n| < \delta \Rightarrow |f(x_m) - f(x_n)| < \varepsilon$.

$\Rightarrow \{f(x_n)\}$ is also a Cauchy Sequence

f is uniformly cont. $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$

x_n Cauchy : $m, n > N \Rightarrow |x_m - x_n| < \delta \Rightarrow |f(x_m) - f(x_n)| < \varepsilon$

Cauchy

Remark : we never have $\{x_n\}$ is Cauchy

Theorem (Heine Cantor)

Let $I = [a, b]$ be closed & bounded interval. if $f : I \to \mathbb{R}$ is continuous then f is uniformly continuous on I.

proof: if not : $\exists \varepsilon_0 > 0\ \forall \delta > 0 .\ \forall n \in \mathbb{N} .\ \exists x_n, y_n \in I .\ |x_n - y_n| < \delta$ but $|f(x_n) - f(y_n)| \ge \varepsilon_0$

Then $\{x_n\} \subseteq I = [a, b]$ $\xrightarrow{\text{B-W}}$ $\exists$ subsequence (x_{n_k}) is convergent

Say $(x_{n_k}) \to \alpha$.

Then $|y_{n_k}| \le 1$ By B-W theorem, $\exists$ subsequence of (y_{n_k}) is convergent.

Say $(y_{n_{k_j}}) \to \beta$.

Subsequence of (x_{n_k}) : $(x_{n_{k_j}})$ also convergent to α.

$|x_{n_{k_j}} - y_{n_{k_j}}| \to |\alpha - \beta|$ as $j \to \infty$

$\Rightarrow \frac{1}{n_{k_j}} > |x_{n_{k_j}} - y_{n_{k_j}}| \to |\alpha - \beta| = 0$

$\Rightarrow \alpha = \beta \in [a, b]$.

$0 < |f(x_{n_{k_j}}) - f(y_{n_{k_j}})| < \varepsilon \overset{\text{for } j}{\Rightarrow} |f(x_{n_{k_j}}) - f(\beta)| \to 0$.

Contradiction

Theorem : continuous extension criterion

$a < b$. $f : (a, b) \to \mathbb{R}$

f is uniformly continuous on $(a, b) \Leftrightarrow$ there is a Continuous extension:

there is Continuous function $g : [a, b] \to \mathbb{R}$. $g(x) = f(x)$. $\forall x \in (a, b)$

proof: Since g is continuous on $[a, b]$. g is uniformly Continuous on $[a, b]$.

$(a, b) \subseteq [a, b]$. g is uniformly cont. on (a, b)

① let $f : (a, b) \to \mathbb{R}$ be uniformly cont. on (a, b).

Need to find a $g : [a, b] \to \mathbb{R}$. cont. on $[a, b]$

(1). Define $g(a)$, $g(b)$:

"⇒" For any Sequence $\{x_n\} \subseteq (a, b)$. $x_n \to a$.

Let x_n is Cauchy. f is uniformly cont. on (a, b).

⇒ $f(x_n)$ is also a cauchy sequence.

$g(a) = \lim_{n\to\infty} f(x_n) = A$?

If $\{y_n\} \in (a, b)$. $y_n \to a$ Claim: $\lim_{n\to\infty} f(y_n) = \lim_{n\to\infty} f(x_n) = K$, so $g(a)$ is well-defined

$|x_n - y_n| < |x_n - a| + |y_n - a| \to 0$.

$\Rightarrow |f(x_n) - f(y_n)| \to 0 \Rightarrow \lim f(x_n) = \lim f(y_n)$

Let $g(a) = \lim_{n\to\infty} f(x_n) = \lim_{x\to a^+} f(x) = A$.

$g(x)$ is continuous at a.

Similarly we define $g(b) = \lim_{n\to\infty} f(x_n)$, where $x_n \in (a, b)$ $x_n \to b$. $\Rightarrow$ g is cont. at b

$g(x) = f(x)$. $\forall x \in (a, b)$

$\Rightarrow g : [a, b] \to \mathbb{R}$ is Continuous

$f : (a, b) \to \mathbb{R}$ is uniformly cont. $\Leftrightarrow$ $\lim_{x\to a^+} f(x) = \alpha$. $\lim_{x\to b^-} f(x) = \beta$

Ex. $f(x) = \frac{x-1}{\log x}$ $x \in (0, 1)$ is uniformly cont. on $(0, 1)$.

proof: $\lim_{x\to 1^-} \frac{x-1}{\log x} = \lim_{x\to 1^-} \frac{x-1}{\ln x \cdot \log_e x} = 1$.

Let $g(x) = \begin{cases} 0, & x = 0, 1 \\ \frac{x-1}{\log x}, & x \in (0, 1) \\ 1, & x = 1 \end{cases}$

$g(x)$ is continuous on $[0, 1] \Rightarrow f$ is uniformly cont. on $(0, 1)$

Exercise 1. Decide which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

a) If $f(x) \to \infty$ as $x \to \infty$ and $g(x) > 0$, then $g(x)/f(x) \to 0$.

b) If $f(x) \to 0$ as $x \to a+$ and $g(x) \ge 2$ for all $x \in \mathbf{R}$, then $g(x)/f(x) \to \infty$.

c) If $f(x) \to \infty$ as $x \to \infty$, then

$$\sin(x^2 + x + 2)/f(x) \to 0.$$

d) If P, Q are polynomials with $\deg(P) \le \deg(Q)$, then there exists $L \in \mathbf{R}$ such that

$$\lim_{x\to-\infty} \frac{P(x)}{Q(x)} = \lim_{x\to+\infty} \frac{P(x)}{Q(x)} = L.$$

(a) $f(x) = x^2$. $g(x) = x^3$. $f(x)/g(x) = 1/x \to 0$

(b) $\forall \varepsilon > 0\ \exists\ \delta = \frac{2}{M}$. $a - \delta < x < a$. $|f(x)| < \frac{2}{M}$

$\forall M \in \mathbb{R}$. $\exists\ g(x)/f(x) \ge 2/f(x) > M$

$a - \delta < x < a$

$\frac{g(x)}{f(x)} \to \infty$ as $x \to a+$

(c). $-1/f(x) \le \sin(x^2 + x + 2)/f(x) \le 1/f(x)$

Since $f(x) \to \infty$ as $x \to \infty$.

$-1/f(x) \to 0$. $1/f(x) \to 0$ as $x \to \infty$.

Sqeeze thm : $\sin(x^2 + x + 2)/f(x) \to 0$ as $x \to \infty$

(d.)

Exercise 2: (Cauchy). Suppose that $f : \mathbf{N} \to \mathbf{R}$. If

$$\lim_{n\to\infty} (f(n+1) - f(n)) = L,$$

prove that

$$\lim_{n\to\infty} \frac{f(n)}{n} = L.$$

$n > N \Rightarrow |f(n+1) - f(n) - L| < \frac{\varepsilon}{2}$

$\left|\frac{f(n)}{n} - L\right| = \left|\frac{f(n) - f(n-1) + f(n-1) - \cdots + f(N+1) - f(N) + f(N)}{n} - L\right|$

$= \left|\frac{f(n) - f(n-1) + \cdots + f(N+1) - f(N) - (N-N)L}{n} + \frac{f(N) + NL}{n}\right|$

$< \frac{n-N}{n} \cdot \frac{\varepsilon}{2} + \left|\frac{f(N) + NL}{n}\right|$

Since $\lim_{n\to\infty} \left|\frac{f(N) + NL}{n}\right| = 0 \Rightarrow \exists N_1 \quad n > N_1 \Rightarrow \left|\frac{f(N) + NL}{n}\right| < \frac{\varepsilon}{2}$

$n > \max(N, N_1) \Rightarrow \left|\frac{f(n)}{n} - L\right| < \frac{n-N}{n} \cdot \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon$

Exercise 3. Decide which of the following statements are true and which are false.

a) If f is continuous on $[a, b]$, then $f([a, b])$ is a closed bounded interval.

b) If f, g are continuous on $[a, b]$, $f(a) < g(a)$ and $f(b) > g(b)$, then there exists $c \in [a, b]$ such that $f(c) = g(c)$.

c) Suppose $f(a) \neq 0$ and f is continuous at a. Then g is continuous at a if and only if fg is continuous at a.

d) Suppose f and g are finite valued on $\mathbf{R}$. If f and $g \circ f$ are continuous, then g is continuous.

(a) Since f is cont. on $[a, b]$. $\exists\ x_m, x_M \in [a, b]$ $f(x_m) = \inf_{x\in[a,b]} f(x)$ $x \in [a, b]$)

$f(x_M) = \sup_{x\in[a,b]} f(x)$ $x \in [a, b]$)

$\Rightarrow f([a, b])$ is bounded

and since $f(x_m), f(x_M) \in f([a, b])$. it is closed

$f(x_m) \le g(x_m) < g(x) < f(x_M) \le f(x)$

$f(x_m) < g(x_m) < g(x) < f(x) < f(x_M)$